

Andrew Boldi

Solving hard science problems with machine learning

📍 San Carlos, CA ✉ andrewjboldi@gmail.com 📞 +1 (650)-590-4181 🔗 andrewboldi 🌐 andrewboldi

Education

UC Berkeley — *BS in Chemical Biology & BA in Computer Science* (GPA: 3.49) *Expected June 2027*

Coursework: Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Computer Architecture, Deep Neural Networks & Computer Vision (Fall '26), Computational Chemistry, Physical Chemistry (Quantum & Statistical Mechanics), Honors Organic Chemistry I/II, Chemical Biology (Biochemistry)

Work Experience

AI Intern — *Soley Therapeutics: Molecular Machine Learning* (Dr. Ankur Gupta) *South San Francisco, CA*
May 2025 - Present

- Full-time summer intern; received return offer and continued projects part-time during the academic year
- Developed new explainability technique for molecular ML models (manuscript in preparation)
- Finetuned hundreds of models on public and proprietary data to predict molecular target candidate profiles (TCPs); built automated benchmarking framework spanning 22 datasets (TDC ADMET group)
- Implemented and benchmarked several architecture ideas for molecular property prediction foundation models
- Automated CDD protocol creation (Selenium + Python), completing a 100+ manual-hour task in 5-6 hours
- Tools used: PyTorch, PyTorch Lightning, Hydra/OmegaConf, CDD Vault, AWS (EC2, S3, CodeBuild)

Research Experience

Summer Research Intern — *Caltech: Stoltz Lab Group* (Dr. Scott Virgil) *Pasadena, CA*
Summers 2022 & 2023

- Completed 16 consecutive steps in a diterpenoid total synthesis; synthesized korormicin & ineleganolide fragments
- Performed computational modeling (ORCA) to determine selectivity in a late-stage stereoinversion
- Developed [StremLigand](#), a catalyst & ligand cataloging application

Assistant Researcher — *Caltech: Center for Catalysis and Chemical Synthesis* *Remote*
July 2020 - May 2022

- Developed remote-automation software for robotic autosamplers (Python, VBA, M68k assembly); used it to screen lanthanide-ligand catalysts for an asymmetric reduction

Projects

Molecular Optical Rotation Prediction with Machine Learning [Trennen](#)

- Predicted rotation direction of optically active molecules with a T-Net (63% accuracy, 50% random baseline); distilled 8M+ molecules to ~27,000 optically active ones via PubChem API, RDKit, and Openbabel

Scenic Navigation Algorithm with Deep Learning [Pathfinder](#)

- Built scenic routing algorithm (modified A*); segmented 250K+ scraped Street View images on Google TPUs

Pacman with Bitcoin Miner [Video Demo](#)

- Top-15 project among ~1400 students in UC Berkeley's data structures course (CS61B); 3000+ lines of Java

Math Competition Training Website [AMC Trainer](#)

- Built training site for math competitions, used by 20K+ users in 50+ countries (Cloudflare Workers, Node.js)

Skills

Programming: Python, Java, Rust, C/C++, JS/CSS/HTML, Svelte, Swift, M68k Assembly, LaTeX

ML & Software: PyTorch, PyTorch Lightning, TensorFlow, RDKit, Hydra/OmegaConf, Selenium, AWS, Git, Slurm, Linux

Chemistry: Multistep organic synthesis, HPLC, LC/MS, NMR, computational modeling (ORCA, DFT)

Languages: English, Armenian, Spanish

Additional

- **Y Combinator AI Startup School (2025):** one of 2000 selected attendees (~20 invited to partner dinner)
- **Piano:** performed at Carnegie Hall and Merkin Hall; 30+ competition recognitions ([YouTube](#))
- **Rubik's cubes:** solved 3x3 in 9.42 seconds, including blindfolded solves; 2x2 through 7x7, Megaminx, Square-1